Low-Resource Nepali Sign Language Alphabet Recognition on NSL23: Comparing RGB Video Models and Hand Landmarks for Webcam Use

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Abstract

Public Nepali Sign Language (NSL) video is scarce, and reported accuracies on other NSL collections cannot be ranked against one another because they use different classes, splits, and sensors. This paper studies isolated alphabet recognition on the public NSL23 release. After auditing the files, the primary sample is 375 original, individually labelled clips from seven signers. Aggregate recordings without timestamps and cropped derivatives are excluded. Two complete-alphabet signers (S1, S2) are held out in turn, giving 277 training clips and 98 test clips per fold. Five models are trained under the same 30-epoch, three-seed protocol: a frozen ImageNet MobileNetV2 classifier on the middle frame, a CNN–BiLSTM, a CNN–Transformer, a multilayer perceptron on MediaPipe hand landmarks, and a bidirectional LSTM on the same landmarks. On 196 pooled unseen-signer clips, RGB accuracy remains $5.8\% \pm 0.8\%$ to $7.0\% \pm 2.1\%$ (chance is 2.0% for 49 classes). Wrist-relative xyz landmarks reach $24.8\% \pm 4.0\%$ with a BiLSTM, mostly on vowels ($66.0\% \pm 4.0\%$ versus $10.0\% \pm 3.9\%$ on consonants). A second matrix on the same folds concatenates 20 inter-joint angles and replaces the softmax head with prototypical nearest-class means [2, 13]. That encoder reaches $59.4\% \pm 3.3\%$ unseen-signer accuracy, with consonants at $61.8\% \pm 5.7\%$. Closed 1-shot enrollment, using the other lighting of the held-out signer as the only support set, raises a cross-entropy BiLSTM to 76.2%–78.2%. A CPU webcam loop with the prototypical encoder sustains 33–42 frames per second on an Apple M4. The result is a stronger practice demo than the xyz softmax baseline. It is still not an interpreter, and it is not a claim of state-of-the-art NSL recognition.

Keywords: Nepali Sign Language; NSL23; signer-independent evaluation; hand landmarks; MediaPipe; prototypical networks; joint angles; webcam; low-resource sign recognition

1 Introduction

A classifier that sees the same person in training and test can use clothing, background, camera, or idiosyncratic articulation. Holding the signer out asks a different question: does the system recognize the sign, or the person who produced it? Large isolated-sign collections treat that distinction as part of the task [7, 12, 17]. NSL does not yet have a public video benchmark of that scale.

NSL23 is a public alphabet video set with 36 consonants and 13 vowels [15, 16]. The release is useful precisely because it is public and licensed for reuse (CC BY 4.0). It is also easy to miscount. The dataset paper reports 1,205 gesture instances in 630 videos. Some files are one labelled letter; others are long aggregate takes; still others are cropped copies of an original take. Treating every file, or every reported instance, as an independent labelled clip inflates the sample and can leak the same performance into both sides of a split.

The practical goal that motivated this work is narrower than a new sequence architecture. The motivating application is an alphabet recognizer that can run from a laptop camera without a GPU. A CNN–Transformer on full RGB frames was a reasonable first candidate: MobileNetV2 is small [10], and Transformers are established for sign sequences [1, 18]. After running that comparison under a frozen signer-held-out protocol, the RGB models sat a few points above chance and the temporal heads reached perfect training accuracy. That is a negative result about representation, not a reason to add more attention layers.

Hand landmarks are a different representation. MediaPipe Hands returns 21 three-dimensional keypoints per detected hand [19]. The vector is small enough for a CPU classifier. It also drops most of the clothing and background that a frozen ImageNet encoder still sees. NSL letter systems have used landmarks on other image collections [11]. Those systems were not evaluated on the original isolated NSL23 clips with S1 and S2 held out.

This paper keeps the audited 375-clip sample and the two signer-held-out folds, then adds landmark classifiers, a follow-on representation study, and a measured webcam loop. The questions are:

1. How many original NSL23 clips support isolated-letter evaluation without inferred segment boundaries?
2. On those folds, do frozen-RGB temporal heads improve over a middle-frame CNN?
3. Do wrist-normalized hand landmarks improve over the same RGB controls?
4. Can a landmark model run as a sliding-window camera demo at interactive frame rates on a laptop CPU?
5. On the same folds, do inter-joint angles and prototypical classification raise unseen-signer accuracy above the xyz softmax BiLSTM?
6. If the other lighting of the held-out signer is available as a 49-way 1-shot support set, how much does that enrollment protocol add?

The answers are descriptive. Only two signers provide a complete 49-class unseen test. Scores from other NSL papers, including word-level Transformers [4] and still-image letter CNNs [11], are not competitors on this exam. Chamachot and Lertniphonphan's angle features and few-shot protocol are a method source, not a leaderboard for NSL23 [2].

2 Related work

2.1 Sign recognition and signer splits

Surveys of deep sign recognition describe a recurring pattern: a spatial encoder, a temporal aggregator, and a dataset-specific split [9]. Continuous systems such as Sign Language Transformers operate over sentence-length sequences and translation objectives [1]. Isolated benchmarks such as MS-ASL, WLASL, and AUTSL instead classify a single sign, often with an official user-independent partition [7, 12, 17]. Those numbers cannot rank an NSL23 model. They do justify reporting signer-held-out scores rather than a random clip shuffle.

2.2 Nepali Sign Language systems

Published NSL systems are mostly private collections. Heuju and Rijal evaluated Transformer and CNN–RNN models on word videos with a 70/30 split; they report 55.9% on 71 words and 89% on a nine-word subset [4]. Shrestha et al. used MediaPipe landmarks and a CNN for letters and fingerspelling on their own images [11]. Those papers show that both attention and landmarks have already been tried for NSL. They do not supply a score on NSL23's isolated original clips. We therefore compare models only against each other on the same held-out files.

NSL23 itself is a dataset paper, not a recognition leaderboard [16]. Collection took place at Sikkim Manipal Institute of Technology in Sikkim, India. Secondary analysis of the public files is not field validation with Deaf NSL users in Nepal.

2.3 Lightweight RGB encoders and pose

MobileNetV2 is a compact ImageNet classifier [10]. Pretrained visual features often transfer, but transfer quality depends on the domain gap [6]. Isolated NSL letters are hand shapes, often nearly static, recorded at variable resolution. A frozen ImageNet encoder is a controlled baseline, not an assumption that those features are appropriate. Pose and landmark pipelines replace the image with keypoints. MediaPipe Hands is one such detector, designed for on-device tracking [19]. Detection can fail under occlusion, blur, or poor lighting; those failure rates belong in the experiment rather than as a reason to avoid the tool.

2.4 Geometric hand features and prototypical classifiers

Wrist-relative coordinates still rotate with the camera and the forearm. Inter-joint angles computed from the same 21 MediaPipe points are invariant to rotation, translation, and isotropic scale. Chamachot and Lertniphonphan use a 20-angle descriptor with a prototypical network on static fingerspelling images from several languages [2]. Prototypical networks classify a query by Euclidean distance to a class mean in an embedding space [13]. We reuse those two ideas on NSL23 video clips, with a BiLSTM encoder over 16 frames, without claiming a result on their static alphabets.

Table 1: Audited NSL23 counts used in this study. "Instances" in the source article are not separately labelled clips.

Quantity	Count
Decodable videos in the release	630
Individually labelled files	607
Aggregate recordings (no timestamps)	23
Cropped or preprocessed files	234
Primary original isolated clips	375
Primary signers	7
Alphabet classes	49
Training clips per fold	277
Test clips per fold (S1 or S2)	98
Seeds × models × folds	$3 \times 5 \times 2 = 30$ runs

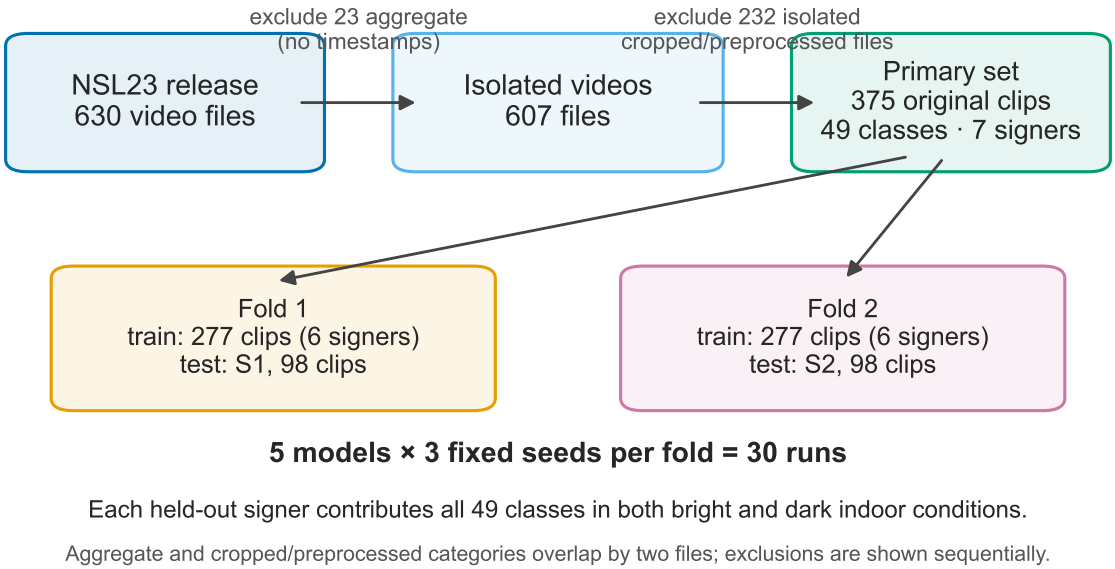


Figure 1: From the NSL23 release to the 30-run matrix. Aggregate files and cropped derivatives are removed before the two signer-held-out folds.

3 Dataset and evaluation protocol

3.1 Release audit

NSL23 version 2 was downloaded from the official Zenodo record [15]. Byte sizes and MD5 digests of the four archives and the dataset-detail document match the record. A file-level manifest retains archive, path, signer token, filename, canonical class, recording descriptors, and exclusion reason.

The audit counts 630 decodable videos: 607 individually labelled files and 23 aggregate recordings. The distribution also contains 234 cropped or preprocessed files. The primary sample is the complement of both exclusions: 375 original, individually labelled clips from S1, S2, S7, S8, S10, S11, and S12 (Figure 1). All 49 alphabet classes appear in that set.

Aggregate recordings were inspected rather than discarded by name. The release does not

provide per-gesture timestamps. Equal-duration bins in published alphabet order drift; at least one sequence appears incomplete or shifted. An equal-duration cut can therefore assign the wrong class while still showing a plausible hand. No candidate bin is treated as ground truth.

Cropped copies are excluded so that a derived rendering of the same take cannot sit in the opposite fold, and so that every prediction points at one original file.

3.2 Class list and folds

The ontology is 36 consonants followed by 13 vowels. Filename aliases are resolved with signer-aware rules because the release uses more than one transliteration. No unknown class is added. No test clip is dropped because its label is missing from training. Each fold contains all 49 labels on both sides of the split.

Only S1 and S2 have original isolated clips for the full alphabet. Fold 1 trains on S2, S7, S8, S10, S11, and S12 and tests on S1. Fold 2 swaps S1 into training and tests on S2. Each test signer contributes one bright and one dark original clip per class. There is no validation set and no early stopping. The epoch-30 checkpoint is evaluated once per run.

The folds are uneven by sign type. S7, S8, S10, S11, and S12 contribute prepared consonants only. Isolated vowels exist only for S1 and S2, so each fold learns every vowel from the other complete-alphabet signer. A correct vowel prediction therefore crosses a one-to-one signer boundary. Consonant training has more signers, but the test person and the unprepared indoor setting remain unseen.

3.3 Frame sampling

Sixteen frames are taken at the centres of equal temporal bins of each clip, resized with aspect ratio preserved, and letterboxed to 224×224. ImageNet channel mean and standard deviation are applied at load time. Training uses brightness and contrast perturbations drawn once per clip and applied to all 16 frames. Horizontal flipping is disabled. The single-frame CNN uses the middle element of the same 16-frame sample.

4 Methods

Figure 2 shows the two pipelines. RGB models share a frozen encoder. Landmark models share MediaPipe Hands and a wrist-relative 64-dimensional frame vector.

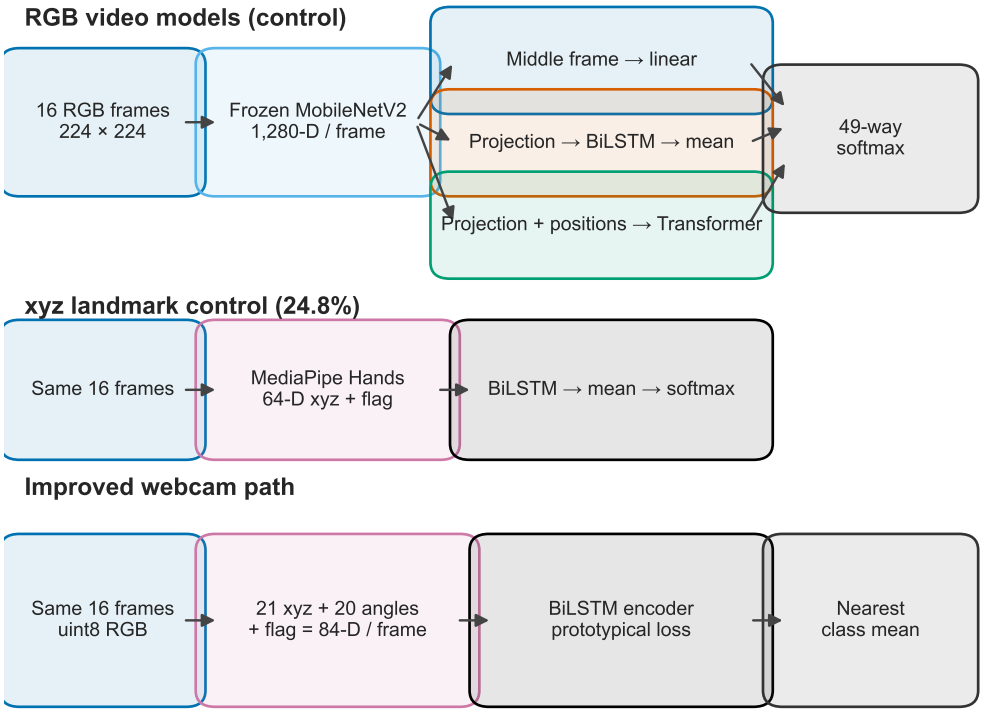
4.1 Frozen RGB models

Let a clip be $\mathbf{X} \in \mathbb{R}^{T\times 3\times 224\times 224}$ with $T = 16$. A frozen MobileNetV2 encoder f_θ with ImageNet-1K V2 weights maps each frame to $\mathbf{z}_t \in \mathbb{R}^{1280}$ [10]. Encoder weights and batch-normalization running statistics are not updated.

The single-frame CNN classifies the middle frame:

$$\mathbf{h} = f_\theta(\mathbf{X}_{\lfloor T/2 \rfloor}), \qquad \hat{\mathbf{y}} = \mathrm{softmax}(W\,\mathrm{Dropout}(\mathbf{h}) + b). \tag{1}$$

It has 62,769 trainable parameters.



The camera demo uses the 84-D prototypical encoder. Closed 1-shot enrollment replaces train means with the other lighting of the test signer.

Figure 2: RGB controls (top), xyz landmark softmax (middle), and the improved webcam path (bottom): 84-D xyz+angle frames and nearest class prototype.

The CNN–BiLSTM projects $\mathbf{z}_t$ to $\mathbf{u}_t \in \mathbb{R}^{256}$, applies a one-layer bidirectional LSTM with 128 units per direction [5], layer-normalizes the outputs, and mean-pools time:

$$\mathbf{u}_t = W_p \mathbf{z}_t + b_p, \qquad [\overrightarrow{\mathbf{h}}_t; \overleftarrow{\mathbf{h}}_t] = \mathrm{LSTM}(\mathbf{u}_{1:T}), \qquad \bar{\mathbf{h}} = \frac{1}{T} \sum_{t=1}^{T} \mathrm{LN}([\overrightarrow{\mathbf{h}}_t; \overleftarrow{\mathbf{h}}_t]). \tag{2}$$

It has 736,305 trainable parameters.

The CNN–Transformer uses the same projection, adds a learned positional table $\mathbf{P} \in \mathbb{R}^{T \times 256}$, and applies two encoder layers with four heads and feed-forward width 512 [18]:

$$\mathbf{H} = \mathrm{Transformer}(\mathbf{U} + \mathbf{P}), \qquad \bar{\mathbf{h}} = \frac{1}{T} \sum_{t=1}^{T} \mathrm{LN}(\mathbf{h}_t). \tag{3}$$

Each head computes $\mathrm{softmax}(QK^\top / \sqrt{d_h})V$ with $d_h = 64$. Encoder dropout is 0 (deterministic scaled-dot-product attention on the training host). Head dropout is 0.2. There is no class token and no causal mask. The model has 1,399,345 trainable parameters.

4.2 Hand-landmark models

MediaPipe Hands [19] is run on the same 16 cached RGB frames (`static_image_mode=True`, complexity 1, at most two hands, detection confidence 0.35). If several hands are returned, the

highest-scoring hand is kept.

Let $\mathbf{l}_i \in \mathbb{R}^3$ be the image-normalized coordinates of landmark $i \in \{0, \ldots, 20\}$, with $i = 0$ the wrist and $i = 9$ the middle-finger metacarpophalangeal joint. Wrist-relative, scale-normalized features are

$$\mathbf{p}_i = \mathbf{l}_i - \mathbf{l}_0, \qquad s = \|\mathbf{p}_9^{(x,y)}\|_2, \qquad \tilde{\mathbf{p}}_i = \frac{\mathbf{p}_i}{\max(s, \varepsilon)}, \tag{4}$$

with $\varepsilon = 10^{-6}$. The frame vector concatenates the 63 coordinates and a detection flag $d_t \in \{0, 1\}$:

$$\mathbf{x}_t = \left[\tilde{\mathbf{p}}_0; \ldots; \tilde{\mathbf{p}}_{20}; d_t\right] \in \mathbb{R}^{64}. \tag{5}$$

A miss yields the zero vector, including $d_t = 0$. No class-conditional imputation is used.

On the 375 primary clips (6,000 frames), Hands detected a hand in 5,868 frames (97.8%). No clip had zero detections. Training adds isotropic Gaussian noise with standard deviation 0.02 to the 63 spatial coordinates; the flag is not noised.

The landmark MLP flattens the clip to $\mathbb{R}^{1024}$ and uses hidden widths 256 and 128 with ReLU and dropout 0.3 (301,617 parameters). The landmark BiLSTM is a one-layer bidirectional LSTM with 64 units per direction, layer normalization, temporal mean pooling, and dropout 0.3 (73,137 parameters):

$$[\overrightarrow{\mathbf{h}}_t; \overleftarrow{\mathbf{h}}_t] = \mathrm{LSTM}(\mathbf{x}_{1:T}), \qquad \bar{\mathbf{h}} = \frac{1}{T}\sum_{t=1}^{T} \mathrm{LN}([\overrightarrow{\mathbf{h}}_t; \overleftarrow{\mathbf{h}}_t]). \tag{6}$$

4.3 Joint angles and prototypical training

The follow-on matrix keeps the same 16-frame MediaPipe cache. From the wrist-relative points $\tilde{\mathbf{p}}_i$, we compute 15 intra-finger flexion angles (three along each of the five rays from the wrist) and five abduction angles at the wrist, following the 20-angle layout of Chamachot and Lertniphonphan [2]. For unit vectors $\mathbf{u}$ and $\mathbf{v}$ from a pivot joint,

$$\phi(\mathbf{u}, \mathbf{v}) = \arccos\big(\mathrm{clip}(\mathbf{u}^\top \mathbf{v}, -1, 1)\big). \tag{7}$$

A miss ($d_t = 0$) yields a zero angle vector. Concatenating the 63 coordinates, the 20 angles, and the detection flag gives

$$\mathbf{x}_t^{\mathrm{raw+ang}} = \left[\tilde{\mathbf{p}}_0; \ldots; \tilde{\mathbf{p}}_{20}; \boldsymbol{\phi}_t; d_t\right] \in \mathbb{R}^{84}. \tag{8}$$

The BiLSTM input width grows from 64 to 84 (83,377 parameters). Cross-entropy training is otherwise unchanged.

Prototypical training uses the same BiLSTM as an encoder g_ψ that maps a clip to $\bar{\mathbf{h}} \in \mathbb{R}^{128}$. Each epoch runs 40 episodes. An episode draws, for every class, five support clips and two query clips from the training fold, with replacement when a class has fewer than seven training clips. Class prototypes are support means. The loss is softmax over Euclidean distances [13]:

$$\mathbf{c}_c = \frac{1}{|S_c|}\sum_{(\mathbf{X}, y) \in S_c} g_\psi(\mathbf{X}), \qquad p(y = c \mid \mathbf{X}) = \frac{\exp(-\|g_\psi(\mathbf{X}) - \mathbf{c}_c\|_2)}{\sum_{c'} \exp(-\|g_\psi(\mathbf{X}) - \mathbf{c}_{c'}\|_2)}. \tag{9}$$

After 40 epochs, test clips are classified by nearest *training*-set prototype. The held-out signer never contributes a prototype in this evaluation. An xyz-only prototypical ablation uses the same encoder and episodes with 64-D frames.

4.4 Cropped-hand CNN and fusion

Full-frame RGB leaves the hand as a small region in a 224×224 letterbox. A follow-on control crops a square around the detected hand from the same 16 cached frames, resizes the crop to 224×224, and trains the frozen MobileNetV2 middle-frame classifier on those crops (62,769 trainable parameters). Fusion averages softmax probabilities from a landmark model and the crop CNN with equal weight, specified before seeing test scores.

4.5 Closed 1-shot enrollment

Each test signer has exactly two original clips per class (bright and dark), so a 5-shot enrollment of the test person is impossible. The honest 1-shot protocol uses the other lighting of all 49 classes from that signer as the support set and the remaining 98 clips as queries: when the query is bright, prototypes are the 49 dark clips, and conversely. Missing classes, which do not occur on S1 or S2, would fall back to the training mean. Replacing only the true class prototype while leaving the other 48 classes as other people's means is not reported as 1-shot accuracy; that setup is too easy.

4.6 Loss and optimization

The five models in the primary matrix use inverse-frequency class-weighted cross entropy without label smoothing. For class counts n_c in a training fold,

$$w_c \propto \frac{1}{n_c}, \qquad \frac{1}{C}\sum_{c=1}^{C} w_c = 1, \qquad \mathcal{L} = -\sum_{c=1}^{C} w_c \, y_c \log \hat{y}_c, \tag{10}$$

with $C = 49$. The weights offset the release's two isolated clips per vowel versus up to seven per consonant.

AdamW uses learning rate 3×10^{-4} and weight decay 10^{-4} [8], with cosine annealing over 30 epochs, batch size 8, and seeds 17, 42, and 73. Deterministic algorithms are enabled. RGB models train on Apple M4 MPS; landmark models train on CPU. The schedule, including the absence of a validation split, was fixed before the RGB matrix and reused for landmarks. Prototypical runs use the same optimizer and seeds for 40 epochs, without class-weighted cross entropy, because the episode already visits every class.

4.7 Webcam loop

The demo does not classify a pre-cut clip. It maintains a deque of the most recent $T = 16$ landmark vectors from a camera or video stream, concatenates the 20 joint angles, runs the prototypical BiLSTM encoder on that window, and assigns the nearest stored training-class mean. Class probabilities are a softmax over negated distances, optionally smoothed with an

exponential moving average

$$\mathbf{s}_t = \alpha \mathbf{s}_{t-1} + (1 - \alpha)\hat{\mathbf{y}}_t, \qquad \alpha = 0.4. \tag{11}$$

Streaming speed is wall-clock frames processed per second, including MediaPipe, not RGB clip throughput after offline decoding.

5 Experiments

The primary matrix is five models, two folds, and three seeds (30 runs). A follow-on matrix on the same folds and seeds trains xyz+angle cross-entropy models, prototypical BiLSTMs (xyz and xyz+angle), and the cropped-hand CNN, then evaluates closed 1-shot enrollment. Each test fold has 98 clips: 49 classes $\times$ {bright, dark}. Seed-level scores pool the two non-overlapping held-out signers (196 unique clips). Architecture summaries are the mean and sample standard deviation of those three seed scores. Top-1 accuracy and macro-F1 are the primary descriptors [14]. Equal class support makes pooled accuracy identical to macro-recall.

Class-stratified clip bootstrap intervals (2,000 replicates) are stored with each run. They describe prediction variability on the observed clips [3]. They do not estimate uncertainty over a population of signers.

Bright/dark and consonant/vowel breakdowns were specified in advance. Condition is not an independently randomized illumination factor. Architecture differences are seed-matched percentage points, not hypothesis tests.

RGB batch-one latency was measured on Apple M4 MPS with device synchronization: model-only forward pass versus original-video decode, preprocess, and inference. Landmark clip latency was measured on CPU: model-only versus MediaPipe on 16 cached frames plus the classifier. The webcam loop was timed on original NSL23 `.MOV` files with the display disabled.

6 Results

6.1 RGB controls

Table 2 and Figure 3 summarize pooled scores. The single-frame CNN reaches $5.8\% \pm 0.8\%$ accuracy and $3.5\% \pm 0.2\%$ macro-F1. CNN–BiLSTM is $7.0\% \pm 2.1\%$ / $4.2\% \pm 2.1\%$. CNN–Transformer is $6.3\% \pm 0.8\%$ / $4.4\% \pm 1.3\%$. All three sit above 2% chance and far below any operational alphabet tutor.

Seed-matched differences among RGB models are within about one to two points (Transformer minus single-frame: $+0.5 \pm 1.0$ accuracy points). Figure 4 shows why a larger gap was easy to expect and did not appear: both temporal RGB heads reach mean training accuracy 1.00 by the last epoch, while test accuracy stays near 6%. The extra capacity memorizes the six training signers.

Table 2: Pooled unseen-signer performance (196 clips/seed; mean ± SD over three seeds). Chance accuracy for 49 classes is 2.0%.

Model	Acc. (%)	Macro-P (%)	Macro-R (%)	Macro-F1 (%)
Single-frame CNN	5.8 ± 0.8	3.6 ± 0.6	5.8 ± 0.8	3.5 ± 0.2
CNN–BiLSTM	7.0 ± 2.1	5.2 ± 5.0	7.0 ± 2.1	4.2 ± 2.1
CNN–Transformer	6.3 ± 0.8	5.7 ± 1.6	6.3 ± 0.8	4.4 ± 1.3
Landmark MLP	18.0 ± 2.1	10.3 ± 1.5	18.0 ± 2.1	11.1 ± 1.8
Landmark BiLSTM	24.8 ± 4.0	17.1 ± 5.6	24.8 ± 4.0	16.8 ± 3.8

Table 3: Held-out signer accuracy and macro-F1 (98 clips/fold; mean ± SD over three seeds).

Model	S1 acc. (%)	S1 F1 (%)	S2 acc. (%)	S2 F1 (%)
Single-frame CNN	6.5 ± 0.6	2.8 ± 0.9	5.1 ± 1.0	2.9 ± 1.3
CNN–BiLSTM	8.2 ± 2.0	5.0 ± 3.2	5.8 ± 2.4	2.0 ± 1.1
CNN–Transformer	5.1 ± 1.0	3.7 ± 1.3	7.5 ± 1.2	3.0 ± 1.4
Landmark MLP	18.7 ± 3.1	11.6 ± 2.5	17.3 ± 1.0	10.1 ± 0.7
Landmark BiLSTM	25.9 ± 2.6	17.4 ± 2.0	23.8 ± 5.6	15.0 ± 4.8

6.2 Landmark models

The landmark MLP reaches 18.0% ± 2.1% accuracy and 11.1% ± 1.8% macro-F1. The landmark BiLSTM reaches 24.8% ± 4.0% and 16.8% ± 3.8%. Relative to the single-frame CNN, the BiLSTM difference is +19.0 ± 3.7 accuracy points and +13.2 ± 3.8 macro-F1 points on the same 196 clips. Relative to the CNN–Transformer it is +18.5 ± 4.6 and +12.4 ± 5.1 points. The BiLSTM also leads the landmark MLP by +6.8 ± 1.9 accuracy points.

Final-epoch training accuracy is 0.37 ± 0.03 for the MLP and 0.58 ± 0.02 for the landmark BiLSTM. Neither saturates. The RGB overfitting pattern does not repeat.

Table 3 splits the two test people. Landmark BiLSTM is 25.9% ± 2.6% on S1 and 23.8% ± 5.6% on S2. The ordering versus RGB is the same in both folds.

6.3 Lighting and sign type

Table 4 and Figure 5 report the pre-specified bright/dark split. Landmark BiLSTM is 27.6% ± 3.5% on bright clips and 22.1% ± 4.6% on dark clips. RGB models stay near 5–7% in both conditions. The dark drop for landmarks is a description of these files, not a controlled lighting study.

Table 5 is the more informative split. RGB vowel accuracy is about 10% for all three image models, including the single-frame CNN, so temporal RGB heads do not explain the vowel subset. Landmark BiLSTM vowel accuracy is 66.0%±4.0%; consonant accuracy is 10.0%±3.9%. The MLP shows the same pattern at a lower level (48.1% vowels, 7.2% consonants). Most of the landmark gain is on the 13 vowels. The 36 consonants remain difficult even after pose normalization.

Figure 6 and Figure 7 show where consonants go. Across the 12 repeated evaluations of each class (two clips × two folds × three seeds), BHA, RA, and TA are never recognized: BHA and RA are always mapped to AH, and TA is always mapped to RI. NA is always mapped to YAN. Those are systematic confusions on this protocol, not rare off-diagonal noise.

Table 4: Descriptive bright versus dark accuracy (98 pooled clips/condition/seed).

Model	Bright acc. (%)	Dark acc. (%)	Bright F1 (%)	Dark F1 (%)
Single-frame CNN	5.8 ± 1.6	5.8 ± 0.6	3.2 ± 1.1	3.0 ± 0.7
CNN–BiLSTM	7.1 ± 1.0	6.8 ± 3.3	3.6 ± 1.6	4.0 ± 3.2
CNN–Transformer	7.1 ± 0.0	5.4 ± 1.6	4.5 ± 0.3	3.0 ± 1.5
Landmark MLP	21.4 ± 2.0	14.6 ± 2.1	12.7 ± 2.5	8.8 ± 1.2
Landmark BiLSTM	27.6 ± 3.5	22.1 ± 4.6	18.4 ± 3.1	14.3 ± 4.1

Table 5: Consonant versus vowel accuracy on the pooled out-of-fold clips.

Model	Cons. acc. (%)	Vowel acc. (%)	Cons. F1 (%)	Vowel F1 (%)
Single-frame CNN	4.2 ± 0.7	10.3 ± 2.2	2.8 ± 0.5	10.9 ± 2.6
CNN–BiLSTM	5.6 ± 1.8	10.9 ± 2.9	4.2 ± 1.8	8.8 ± 3.5
CNN–Transformer	4.9 ± 0.7	10.3 ± 2.9	4.0 ± 0.8	9.5 ± 2.6
Landmark MLP	7.2 ± 2.4	48.1 ± 1.9	5.6 ± 2.5	45.0 ± 2.8
Landmark BiLSTM	10.0 ± 3.9	66.0 ± 4.0	8.4 ± 4.0	64.2 ± 2.7

6.4 Joint angles, prototypical classification, and crops

Table 7 and Figure 8 report the follow-on matrix on the same 196 clips per seed. Concatenating the 20 angles under cross entropy raises the BiLSTM from 24.8% ± 4.0% to 37.9% ± 2.4%. Consonant accuracy moves from 10.0% to 28.7% ± 4.5%; vowels stay near 63.5%. The angle MLP does not beat the xyz BiLSTM.

Prototypical training is the larger change. An xyz prototypical BiLSTM reaches 55.8% ± 1.8%. Adding angles to that encoder reaches 59.4% ± 3.3% accuracy and 54.7% ± 4.2% macro-F1. Consonants are 61.8% ± 5.7%; vowels are 52.6% ± 4.8%. The softmax xyz model was a vowel specialist. The prototypical encoder trades some of that vowel rate for a usable consonant rate. Fold means are 63.3% ± 3.7% on S1 and 55.4% ± 3.3% on S2.

The cropped-hand CNN is 19.2% ± 1.3%. Cropping helps relative to full-frame RGB (5.8%–7.0%) and does not catch landmarks. Equal-weight softmax fusion of the prototypical encoder and the crop CNN is 59.7% ± 3.6%, within noise of the prototypical encoder alone. Fusion of the angle BiLSTM and the crop CNN is about 41.5%, a few points above the BiLSTM and still far below prototypical classification. Figure 9 is the corresponding confusion matrix.

6.5 Closed 1-shot enrollment

Table 8 and Figure 10 separate the hard exam (train prototypes only) from a tutor setting in which the system has seen the test signer's other lighting of every letter. Closed 1-shot accuracy is 76.2% ± 4.2% for the xyz BiLSTM, 78.2% ± 2.8% for the angle BiLSTM, and 73.0% ± 1.5% for the prototypical encoder. Enrollment helps the weaker softmax models more. It does not replace the unseen-signer number: a new user who has not recorded a dark and a bright take of all 49 letters still faces the 59.4% prototypical exam.

Table 6: Batch-one clip latency. RGB end-to-end includes original-video decode on Apple M4 MPS. Landmark end-to-end is MediaPipe Hands on 16 cached frames plus the CPU classifier. Streaming FPS is reported separately.

Model	Device	Model-only (ms)	End-to-end (ms)	Clips/s
Single-frame CNN	MPS	4.0 (4.2)	939.0 (2097.8)	1.06
CNN–BiLSTM	MPS	49.5 (50.2)	962.9 (2126.2)	1.04
CNN–Transformer	MPS	49.7 (50.1)	970.9 (2162.6)	1.03
Landmark BiLSTM	CPU	0.3 (0.5)	345.7 (441.2)	2.89

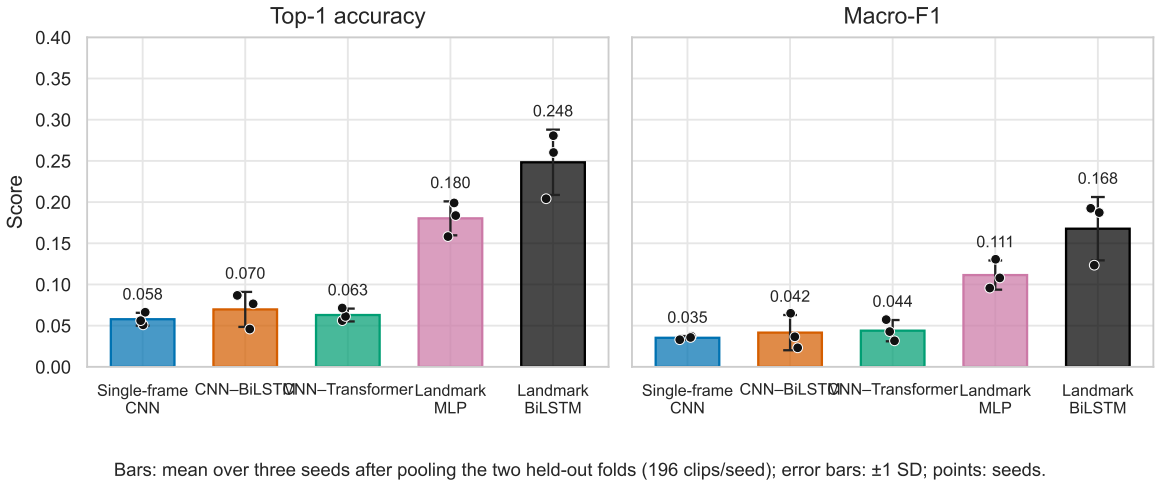


Figure 3: Pooled unseen-signer accuracy and macro-F1. Bars are means over three seeds; error bars are one sample standard deviation; points are seeds.

6.6 Runtime and webcam

Table 6 separates network time from pipeline time. On MPS, RGB model-only medians are 4.0 ms (single-frame) and about 50 ms (temporal heads). End-to-end RGB clip processing is about 0.94–0.97 s because decoding dominates. That measurement is not a camera frame rate.

On CPU, the landmark BiLSTM forward pass is 0.3 ms per 16-frame clip (0.5 ms at the 95th percentile). Running MediaPipe on those 16 cached frames plus the classifier takes a median 346 ms per clip. The webcam loop is cheaper per frame because Hands runs in video mode on one frame at a time and the LSTM is applied to a rolling window.

On original NSL23 videos, with the prototypical xyz+angle encoder trained without the signer in the file, the loop processed 90 frames of S1's bright isolated `A` in 2.27 s (39.6 FPS). The argmax after EMA smoothing was `A`. The earlier xyz softmax loop on the same clip was 39.7 FPS; nearest-prototype classification does not change the MediaPipe-dominated budget. In three xyz-softmax streams reported previously (S1 bright, S1 dark, S2 bright `A`), wall-clock rates were 33–42 FPS. Those measurements are a speed check on one vowel, not a streaming accuracy estimate over the alphabet.

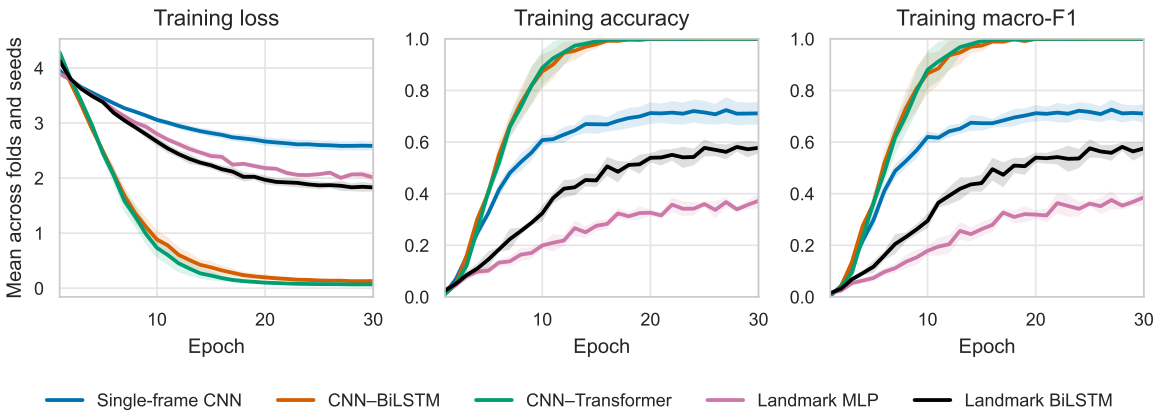


Figure 4: Training loss, accuracy, and macro-F1 (mean ± SD over folds and seeds). RGB temporal heads fit the training clips; landmark models do not reach training accuracy 1.

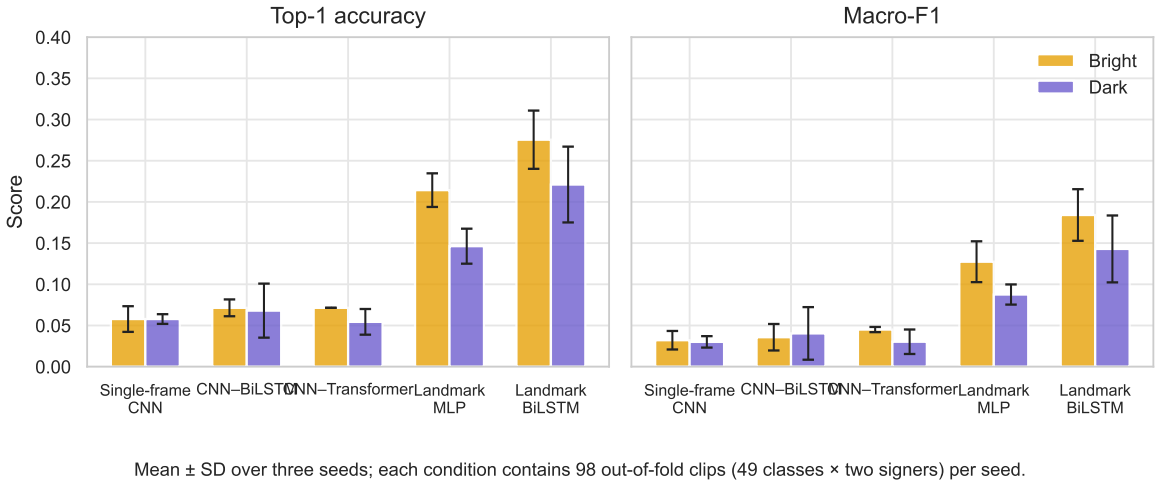


Figure 5: Bright versus dark accuracy and macro-F1. Each condition has 98 out-of-fold clips per seed.

7 Discussion

The RGB comparison answers the original architectural question. With a frozen ImageNet encoder, 16 letterboxed frames, and this isolated subset, a BiLSTM or Transformer head does not produce a reliable unseen-signer gain. Perfect training accuracy and $\approx 6\%$ test accuracy point to a representation failure. Many NSL23 consonant classes are held shapes. ImageNet features, even from MobileNetV2, are a weak description of those shapes when the test person changes.

Landmarks change the input. Coordinates ignore most of the background, and wrist-relative scaling removes some of the absolute position and size of the hand in the frame. Under softmax cross entropy, the model that then works best is a 73k-parameter BiLSTM on 64-dimensional pose vectors (24.8%). That is the frozen control. The webcam path that follows uses the same LSTM width with 84-D frames and nearest-class means (83k parameters), not a Transformer on images.

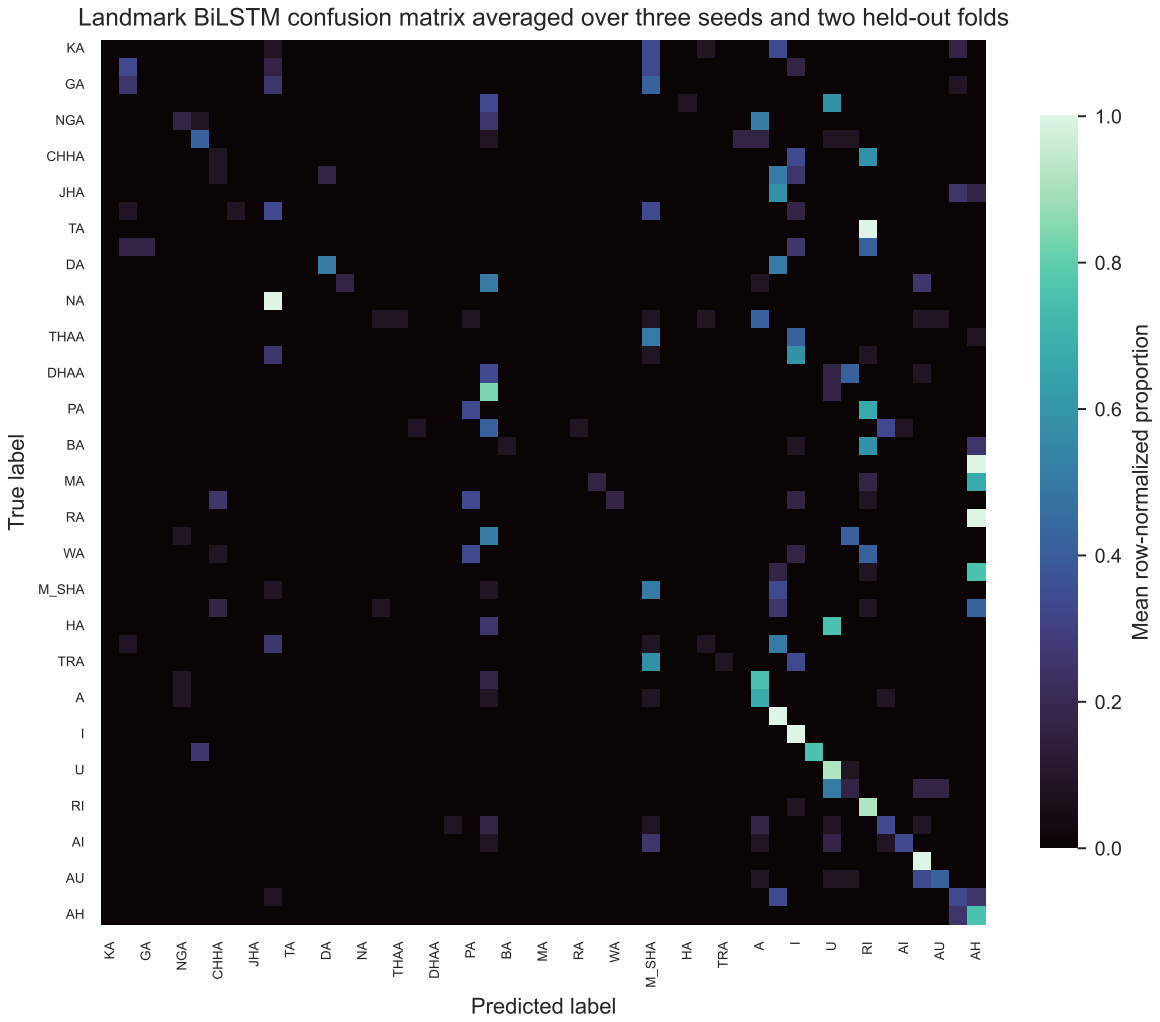


Figure 6: Landmark BiLSTM confusion, row-normalized and averaged over two folds and three seeds. Several consonants collapse onto a few vowel-like predictions.

The vowel/consonant split on the xyz softmax BiLSTM limits how far the 24.8% number should be pushed. Vowels in this release are few, exist for only two isolated signers, and are more separable in landmark space. Consonants are the bulk of the alphabet and were near 10% for that model. Several confusions were complete across seeds.

The follow-on matrix changes the product reading. Prototypical classification, not a deeper RGB network, is what moves consonants. Unseen-signer accuracy of 59.4% is still two errors in five, and it is not translation. It is no longer a system that is almost always wrong on the 36 consonants. Closed 1-shot enrollment at 76–78% is the number that belongs to a tutor that has already seen the learner's other lighting of each letter. Those two settings should not be merged in a demo caption.

Cropping the hand for a frozen ImageNet CNN does not close the gap. The failure of full-frame RGB was not only that the hand was small; the frozen encoder remains a weak description of NSL letter shapes when the person changes.

Other published percentages stay in their own rows. A 90% still-image result on a private or Kaggle letter set is not a loss for this paper, 59.4% on NSL23 is not a win over a 71-word video

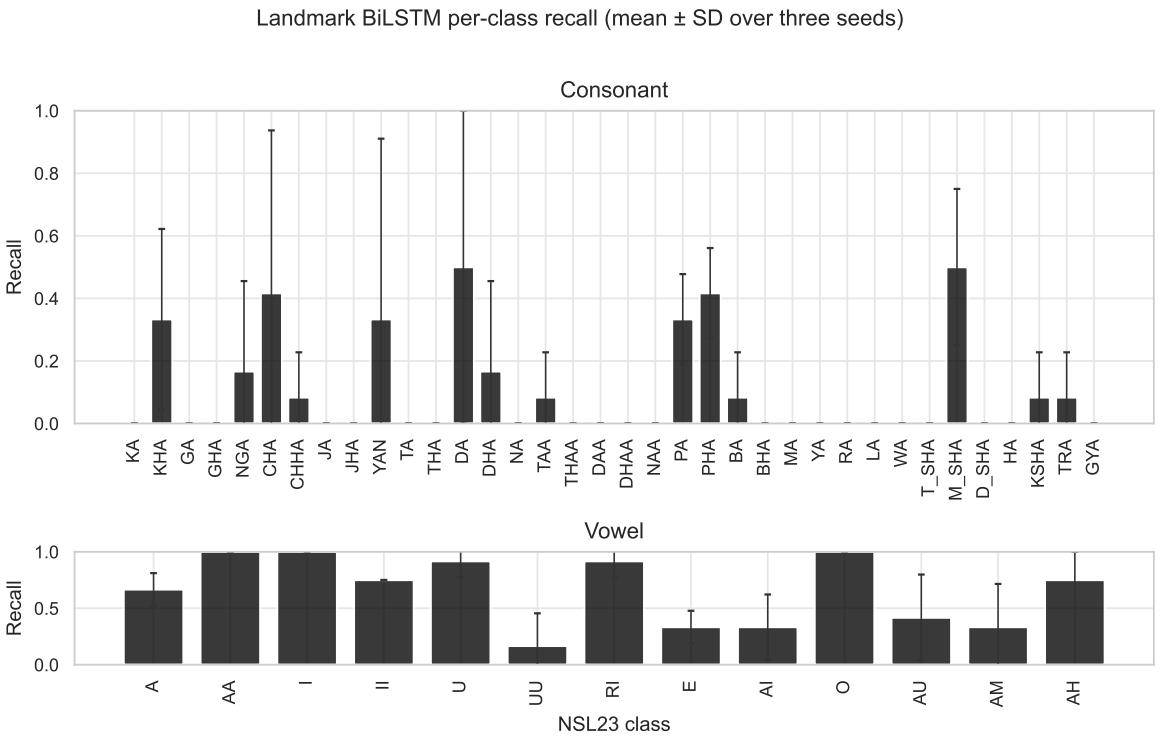


Figure 7: Per-class recall of the landmark BiLSTM (mean ± SD over seeds). Vowels occupy the lower panel.

Table 7: Follow-on unseen-signer accuracy on the same S1/S2 folds (196 clips/seed; mean ± SD over three seeds). The xyz BiLSTM row is the frozen 30-run control.

Model	Acc. (%)	Macro-F1 (%)	Cons. (%)	Vow. (%)
xyz BiLSTM (CE)	24.8 ± 4.0	16.8 ± 3.8	10.0 ± 3.9	66.0 ± 4.0
Hand-crop CNN	19.2 ± 1.3	15.9 ± 0.7	15.5 ± 0.8	29.5 ± 2.9
xyz+angle MLP (CE)	22.4 ± 3.1	14.6 ± 2.9	13.7 ± 3.6	46.8 ± 2.9
xyz+angle BiLSTM (CE)	37.9 ± 2.4	30.1 ± 2.6	28.7 ± 4.5	63.5 ± 3.8
xyz Proto-BiLSTM	55.8 ± 1.8	50.6 ± 2.4	58.3 ± 2.5	48.7 ± 5.9
xyz+angle Proto-BiLSTM	59.4 ± 3.3	54.7 ± 4.2	61.8 ± 5.7	52.6 ± 4.8

study, and the 20-angle descriptor is not a reproduction of Chamachot and Lertniphonphan's static few-shot numbers [2]. Different labels, different files, different splits.

8 Limitations

The complete-alphabet unseen test uses two people. Clip bootstrap intervals do not repair that. Prepared-consonant signers never appear as unseen full-alphabet testers. Outdoor and real-world files in the release are not in the primary sample. Continuous signing, fingerspelling in sentences, and two-handed or body-relative signs are out of scope.

MediaPipe is a third-party detector. Landmark quality is not independent of lighting, blur, skin tone, or camera. A 97.8% frame detection rate on these cached frames does not guarantee the same rate on an unseen webcam.

The RGB encoder was frozen. Fine-tuning MobileNetV2, or pretraining on a large still-image

Table 8: Closed 1-shot enrollment: the other lighting of the held-out signer supplies one support clip per class (98 queries/fold). Mean ± SD over three seeds after pooling S1 and S2.

Encoder	Unseen signer (%)	Closed 1-shot (%)
xyz BiLSTM (CE)	24.8 ± 4.0	76.2 ± 4.2
xyz+angle BiLSTM (CE)	37.9 ± 2.4	78.2 ± 2.8
xyz+angle Proto-BiLSTM	59.4 ± 3.3	73.0 ± 1.5

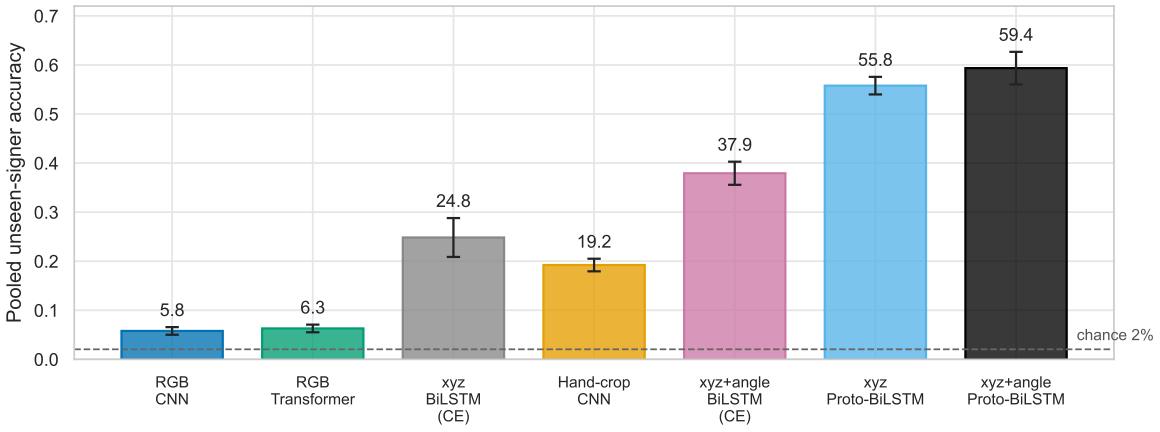


Figure 8: Unseen-signer accuracy after pooling S1 and S2. The xyz BiLSTM is the frozen 30-run control; the remaining bars are the follow-on matrix.

letter set, might raise RGB or crop accuracy. Poudel's Kaggle still-image NSL letters were not used here; this environment had no Kaggle credentials, and that collection is 36 consonants rather than the 49 NSL23 classes. Hyperparameters were not searched after seeing test scores; they may be suboptimal.

Closed 1-shot enrollment uses the other lighting of the same recording session, not a second day and not a live enrollment UI. With two clips per class per signer, 5-shot enrollment of the test person cannot be run honestly.

The webcam measurement uses NSL23 files as a stream, not a live user study. FPS is hardware-specific (Apple M4, MediaPipe 0.10.14, CPU). Interactive speed is not the same as recognition quality.

9 Conclusion

On audited original isolated NSL23 clips, with S1 and S2 held out, frozen RGB video models remain near 6–7% accuracy. Wrist-normalized MediaPipe landmarks raise that to 24.8% ± 4.0% with a softmax BiLSTM, mostly by separating vowels. Concatenating 20 joint angles and training the same encoder with prototypical loss reaches 59.4% ± 3.3% unseen-signer accuracy, with consonants at 61.8%. Closed 1-shot enrollment on the other lighting of the test signer is 76–78% for the cross-entropy BiLSTMs. A CPU webcam loop using the prototypical encoder runs at 33–42 FPS on an Apple M4. The public files, protocol, predictions, and demo script are released so later work can replace the detector, unfreeze the RGB encoder, or change the enrollment UI

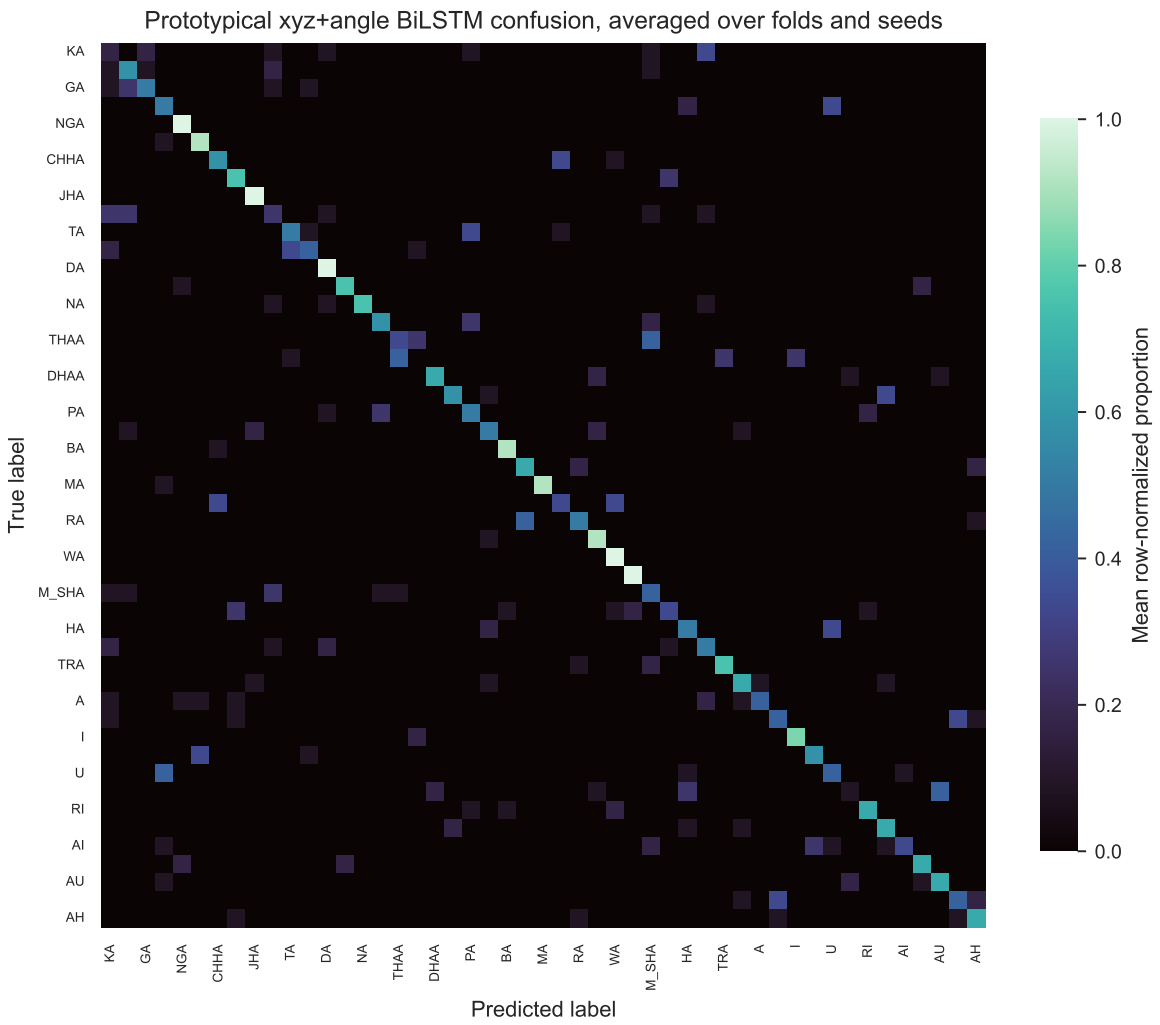


Figure 9: Prototypical xyz+angle BiLSTM confusion, row-normalized and averaged over two folds and three seeds.

without repeating the sample-definition mistakes that make NSL23 easy to overcount.

Data availability

NSL23 videos are distributed by Sunuwar, Borah, and Kharga under CC BY 4.0 [15, 16]. This repository does not republish the archives. Code, fold manifests, landmark caches, checkpoints, and tabulated results are at `https://github.com/twasanon/nsl23-low-resource-alphabet`.

Acknowledgements

The recordings were collected and released by the NSL23 authors. This paper is a secondary analysis of those files.

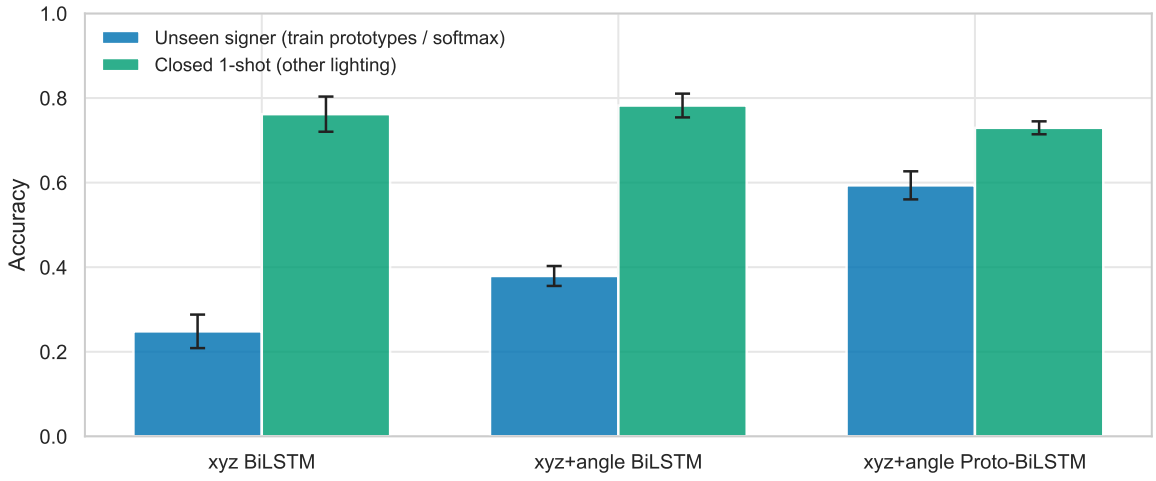


Figure 10: Unseen-signer classification versus closed 1-shot enrollment on the other lighting of the held-out signer.

References

[1] Necati Cihan Camgoz, Oscar Koller, Simon Hadfield, and Richard Bowden. Sign language transformers: Joint end-to-end sign language recognition and translation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 10023–10033, 2020. doi: 10.1109/CVPR42600.2020.01004.

[2] Chayanin Chamachot and Kanokphan Lertniphonphan. Geometry-aware metric learning for cross-lingual few-shot sign language recognition on static hand keypoints. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*, pages 10481–10489, 2026.

[3] Bradley Efron and Robert J. Tibshirani. *An Introduction to the Bootstrap*. Chapman and Hall, 1993.

[4] Shristi Heuju and Birodh Rijal. Word level Nepali Sign Language recognition using transformer. In *Proceedings of the 12th IOE Graduate Conference*, pages 883–890, 2022.

[5] Sepp Hochreiter and Jürgen Schmidhuber. Long short-term memory. *Neural Computation*, 9(8):1735–1780, 1997.

[6] Simon Kornblith, Jonathon Shlens, and Quoc V. Le. Do better ImageNet models transfer better? In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 2661–2671, 2019.

[7] Dongxu Li, Cristian Rodriguez Opazo, Xin Yu, and Hongdong Li. Word-level deep sign language recognition from video: A new large-scale dataset and methods comparison. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, pages 1459–1469, 2020. doi: 10.1109/WACV45572.2020.9093512.

[8] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations*, 2019.

[9] Razieh Rastgoo, Kourosh Kiani, and Sergio Escalera. Sign language recognition: A deep survey. *Expert Systems with Applications*, 164:113794, 2021.

[10] Mark Sandler, Andrew Howard, Menglong Zhu, Andrey Zhmoginov, and Liang-Chieh Chen. MobileNetV2: Inverted residuals and linear bottlenecks. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 4510–4520, 2018.

[11] Guptaraj Shrestha, Nishan Thing, Prajita Dhakal, Prasanga Dahal, Pukar Karki, and Manoj Kumar Guragai. Nepali Sign Language letter detection and finger spelling using MediaPipe and CNN. In *Proceedings of the 15th IOE Graduate Conference*, pages 166–170, 2024.

[12] Ozge Mercanoglu Sincan and Hacer Yalim Keles. AUTSL: A large scale multi-modal Turkish Sign Language dataset and baseline methods. *IEEE Access*, 8:181340–181355, 2020.

[13] Jake Snell, Kevin Swersky, and Richard Zemel. Prototypical networks for few-shot learning. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

[14] Marina Sokolova and Guy Lapalme. A systematic analysis of performance measures for classification tasks. *Information Processing and Management*, 45(4):427–437, 2009.

[15] J. Sunuwar, S. Borah, and A. Kharga. NSL23 dataset for alphabets of Nepali Sign Language. Zenodo, 2023. Version 2, CC BY 4.0.

[16] J. Sunuwar, S. Borah, and A. Kharga. NSL23 dataset for alphabets of Nepali Sign Language. *Data in Brief*, 53:110080, 2024. doi: 10.1016/j.dib.2024.110080.

[17] Hamid Reza Vaezi Joze and Oscar Koller. MS-ASL: A large-scale data set and benchmark for understanding American Sign Language. In *Proceedings of the British Machine Vision Conference*, 2019.

[18] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

[19] Fan Zhang, Valentin Bazarevsky, Andrey Vakunov, Andrei Tkachenka, George Sung, Chuo-Ling Chang, and Matthias Grundmann. MediaPipe hands: On-device real-time hand tracking. In *CVPR Workshop on Computer Vision for Augmented and Virtual Reality*, 2020.